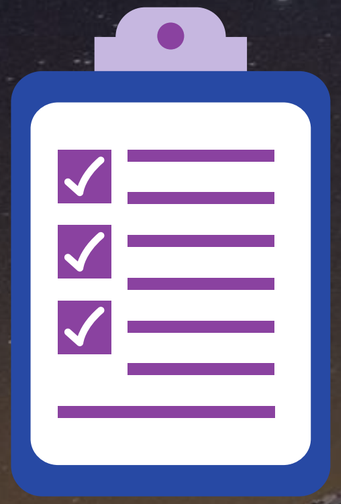
TRAINING FOR MOUNT SHASTA



MAXIMIZE YOUR CHANCES OF SUMMIT SUCCESS

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Facts, Folklore, and Fitness; get ready to climb Northern California's biggest peak!

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03 | Training

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04 | Support

Need additional support? Sign up for training program! We are here to help you on your fitness journey.

MOUNT SHASTA

“When I first caught sight of it over the braided folds of the Sacramento Valley, I was fifty miles away and afoot, alone and weary. Yet all my blood turned to wine, and I have not been weary since.” ~ John Muir, 1874

Rising over 11,000 vertical feet from the valley below, Mount Shasta looms over the vastness of rural Northern California, captivating its visitors and native inhabitants for thousands of years. First climbed in 1854, Mount Shasta has a rich history of exploration, mountain adventures, and spiritual folklore. Prone to extreme weather and massive avalanches, Mount Shasta holds the record for deepest snowfall in a single storm of over 15 feet. Despite the 1 in 4 chance that the dormant volcano may erupt in your lifetime, the lure of Mount Shasta continues to captivate and mystify over 15,000 climbers each year.

The purpose of this training guide is to better prepare you for the physical and mental demands of climbing Mount Shasta. This undertaking should not be taken lightly - although considered a "beginner's mountain", climbing Mount Shasta exposes one to a multitude of objective hazards and has claimed the lives of many over the years. Showing up well-trained can significantly increase your margin of safety, the overall enjoyment of your climb, and of course, your odds of reaching the summit. For additional questions about training for your climb, please reach out to Shasta Mountain Guides.



SHASTA BY THE NUMBERS

14,179 FEET ABOVE SEA LEVEL, THE SECOND TALLEST CASCADE VOLCANO

10,000 MINIMUM NUMBER OF CALORIES YOU WILL BURN ON A 3-DAY TRIP

7,000 VERTICAL FEET OF ELEVATION GAIN FROM EVERY TRAILHEAD

50 PERCENT CHANCE OF REACHING THE SUMMIT, 80% WITH A GUIDE!

45 POUND BACKPACK - THE MINIMUM WEIGHT YOU WILL CARRY TO CAMP

24 AVERAGE NUMBER OF HOURS SPENT MOVING ON A 3-DAY CLIMB

MOUNTAIN

VARIABLES

These variables are out of your control; regardless of your level of experience or training!

Altitude

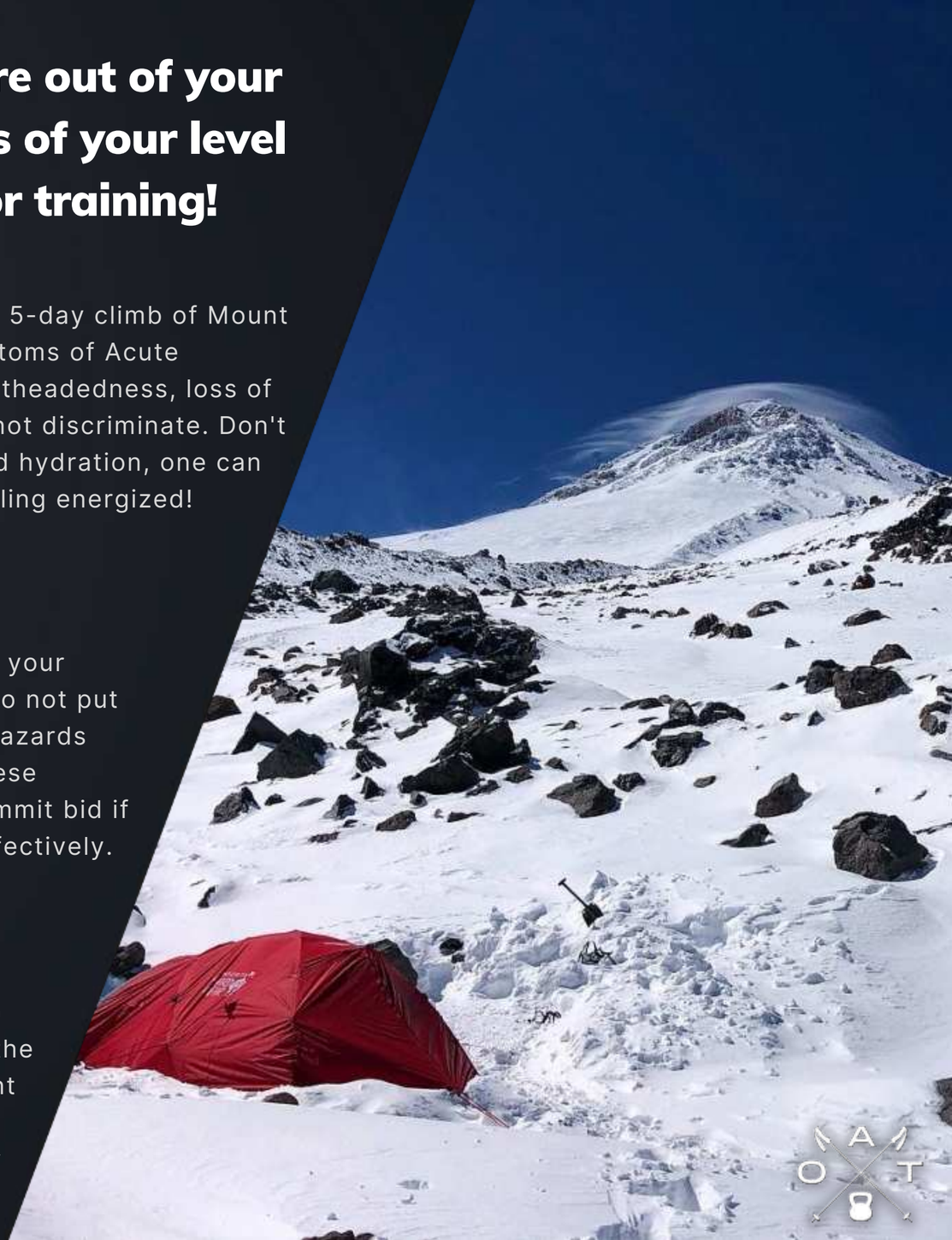
To set the story straight, you will NOT acclimatize during your 3 to 5-day climb of Mount Shasta. Approximately a third of all climbers experience mild symptoms of Acute Mountain Sickness (AMS) including headache, nausea, fatigue, lightheadedness, loss of appetite, and difficulty sleeping. AMS strikes at random and does not discriminate. Don't let this discourage you - with proper pacing, fueling strategies, and hydration, one can safely reach high altitude and return to camp in good style and feeling energized!

Objective Hazard

You have hired a guide to keep you safe on the mountain. As such, your guides will exercise the highest level mountain safety protocol as to not put the team at an unnecessary level of risk. That being said, certain hazards such as rockfall, avalanche, or falls may be unavoidable. Any of these identified hazards may be enough to forfeit an opportunity of a summit bid if your guides deem the risk of these hazards too high to mitigate effectively. Always respect and listen to your guides!

Weather

Weather in the mountains is unpredictable, and Mount Shasta is no exception! Extreme temperatures, wind, and precipitation can put the summit out of reach on any day of the year. Preparing with the right equipment can go a long way towards your success. The weather you are dealt on the week of your summit climb may be out of your control, but there are a lot of factors you can control!



CLIMBER

VARIABLES

These variables are within your control and will significantly impact your chances of summit success!

Physical Fitness

Perhaps the single most important variable within your control is your physical conditioning when you arrive for your climb. Although intensive training does not by any means guarantee a summit, it will significantly increase your chances of success and enjoying your experience on the mountain. A well-rounded mountain athlete should have excellent cardiovascular endurance, joint stability, balance, muscular endurance, power, and mobility. More on all of these topics soon!

Proper Planning

Being prepared for your climb is much more than putting in hours of physical training. Knowing your gear and how to use it goes a long way towards an efficient and again, enjoyable climb. It may be wise to purchase any new gear or clothing well ahead of your climb to practice using it in the field. Watching tutorial videos, following weather patterns, and reading about mountaineering skills may also prove to be valuable components of planning ahead.

Positive Mindset

Often overlooked, one's mental state and overall feelings towards the climb have significant impact on their chances of a successful and enjoyable experience. In the words of best-selling author Jack Canfield, "Whether you believe you can or believe you can't, your thoughts become a self-fulfilling prophecy. It's your choice to believe or not." Words of affirmation are a powerful tool. Trust in your training and always try your best!



APPROACH TO TRAINING

Incorporating these 5 Training Pillars will take your mountain fitness to new heights!

It's important to note that "success" comes in many different forms and does not only pertain to reaching a geographical summit. Mental and emotional summits are often times even more fulfilling then reaching an arbitrary location on a map.

That being said, showing up physically and mentally prepared will increase your likelihood of success. Whether it's overcoming mental or emotional challenges, excelling in the face of adversity, or reaching a new high point, diligent training can be a major catalyst of growth.

Following the **5 Training Pillars** explained in this document will set you up for "success"!

SPORT SPECIFIC TRAINING 1

No training, no matter how rigorous, can replace the benefits of sport-specific training. In the case of climbing a volcano, that means logging a vast majority of your training hours walking up AND down hills!

CARDIO 2

All endurance activities require strong cardiovascular fitness, and intentional training will pay dividends during long days on the mountain. Most of your cardio training should come in the form of aerobic exercise.

STABILITY & STRENGTH 3

Stability and Strength Training contribute greatly to your ability to perform repetitive muscle contractions for hours on end. Resistance training prepares the muscles and joints for the demands of climbing, and reduces risk of injury.

MOBILITY 4

Flexibility and Mobility Training is a critical component of injury prevention and movement pattern optimization. As you ramp up your training load, time spent mobilizing and recovering will enhance your training results.

NUTRITION 5

Nutrition can make or break your training program. A balanced diet of primarily plant-based, whole food sources will enhance your workouts, improve recovery times, and allow you to pursue your favorite hobbies for decades to come!

SPORT-SPECIFIC TRAINING



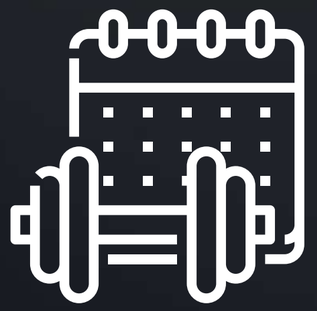
Carry a Pack and Hike!

The Principle of Specific Adaptation to Imposed Demand (SAID) states that the body will adapt according the stress placed upon it. In the case of climbing Mount Shasta, we want our body to adapt to the stimulus it will face on the climb, ahead of the experience. That means doing our best to mimic the motion, repetition and duration the best we can the weeks and months prior. In short, you must start racking up hours of time spent on feet, moving up and down hills, while carrying a weighted pack.



Work With What You Have

For most, it may be impossible to spend hours each week logging long days on the trail; but that does not mean you can't reach the summit! Be resourceful and creative! Find a set of stairs in a nearby park, office building, or use a stair machine at the gym. If you have the opportunity to spend time in mountainous regions, take full advantage of local trails and log some long hikes with significant elevation gain!

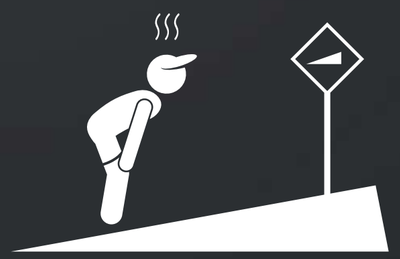


Stay Consistent

Adaptation only comes from persistent effort at regular frequencies. A few training hikes in the weeks before your climb may not be enough. It takes weeks or months of steadily increasing your hiking and overall training load to reach the desired adaption you need to climb with confidence. Set a training schedule, hold yourself accountable, and keep your goals in mind to stay motivated on your journey!



CARDIOVASCULAR TRAINING



Uphill Movement

Tying back into sport-specific training, performing a majority of your cardiovascular training during uphill movement will expedite results. Not only will hill training improve your muscular endurance in the legs, it will help facilitate incredible cardiovascular gains, both aerobically and anaerobically. Keep in mind, hills are HARD! Whether hiking or running, it can take a long time to feel comfortable moving uphill at a moderate pace with a heavy pack. Be kind to yourself and enjoy the process of growth!



Heart Rate Zone Training

One of the most effective ways to improve your cardiovascular fitness is to practice intentional Heart Rate Zone Training. In short, aim to spend most of your time focusing on low intensity, long duration cardio, with only 20-35% of your total training coming from moderate to high intensity bouts. A heart rate monitor is an effective tool to gauge, but so is your breathing! Review the following slide for more details on HR Zone Training.



More Than Just Training

Cardiovascular training can help us achieve results far beyond just reaching a summit. In fact, one study showed that cognitive decline was nearly twice as likely in adults who did not get at least 150 minutes of cardiovascular exercises each week. Other benefits include improved immune system function, improved mood, reduced risk of chronic disease, and much more!



HR max = 220 - your age

HEART RATE ZONES

ANAEROBIC	Zone 5 96-100% HR MAX	Maximal Exertion. Reserved for extreme effots in races or emergency situations. Less than 1% of total training.
ANAEROBIC	Zone 4 90-95% HR MAX	High Intensity. Difficult to maintain for more than a few minutes at a time. Less than 10% of total training.
ANAEROBIC	Zone 3 80-89% HR MAX	Moderate intensity. Switching to Anaerobic System. Helps improve anaerobic threshold. 15-30% of total training.
AEROBIC	Zone 2 71-79% HR MAX	Light Intensity. Slightly above aerobic base. Often hard to differentiate from Zone 1. 10-20% of total training.
AEROBIC	Zone 1 50-70% HR MAX	Low Intensity. Contributes to building strong aerobic base. Foundation for long efforts. 50-70% of total training.

12-WEEK PERIODIZED HIKING PROGRAM



Week	Mileage	Elevation Gain	Pack Weight
Week 1/2:	5-10 miles	1,000-2,000'	20-25 lbs
Week 3/4:	7-12 miles	2,000-3,000'	25-30 lbs
Week 5/6:	7-12 miles	3,000-4,000'	30-35 lbs
Week 7/8:	12-15 miles	4,000-5,000'	35-40 lbs
Week 9/10:	12-15 miles	6,000-7,000'	40-45 lbs
Week 11:	13-17 miles	5,000-7,000'	45-50 lbs
Week 12:	Active Recovery and Light Aerobic Workouts		

This template is an example of a periodized approach to scaling up your hiking load in the weeks leading up to your climb. Note these are WEEKLY totals.

Every BODY is different and this will look very different person to person. One may need more or less time to build up to these numbers based on currently level of fitness, age, history of injuries, and time. Start small, and slowly increase over time.

Regardless of where you start your journey, remember that consistency really is the key to all success! As Lao Tzu says "The journey of a thousand miles begins with a single step."

STRENGTH & STABILITY TRAINING



Balance & Stability

Often overlooked in training programs, joint stability and balance are critical for the outdoor athlete. Walking on steep slopes and uneven terrain require excellent body awareness and spacial proprioception to move safely and efficiently. Balance and stability adaptation can be achieved by utilizing techniques such as single leg exercises, plyometrics, and exercise tools that create unstable environments. Stability Training will optimize joint movement and enhance your overall training results.



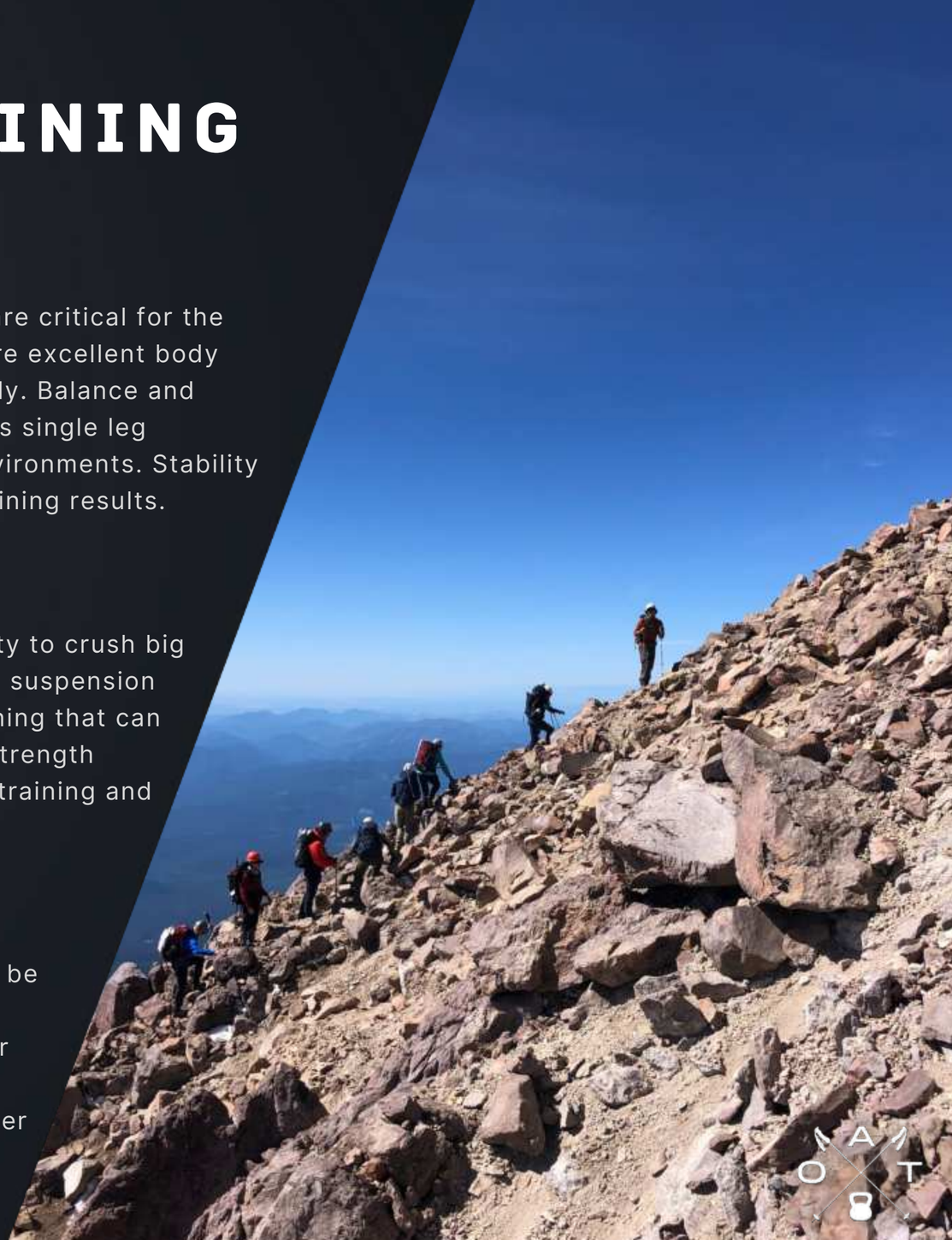
Strength Endurance

Enhancing muscular endurance will significantly improve your ability to crush big days in the mountains! Bodyweight exercises, free weights, bands, suspension trainers, and even uphill movement are all forms of resistance training that can stimulate adaptation in the musculoskeletal system. A structured strength training program is an excellent way to supplement sport-specific training and take your overall fitness to new heights.



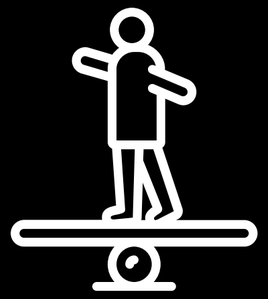
Proper Biomechanics

The importance of excellent form and proper biomechanics cannot be overemphasized. Failure to maintain posture and proper form can result in chronic or acute injury, setting you further away from your goals. If you have little or no experience with structured strength training, seek professional guidance from a certified personal trainer to optimize results and reduce risk of injury.



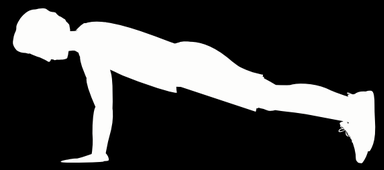
STABILIZATION

NEVER ADD STRENGTH TO DYSFUNCTION!



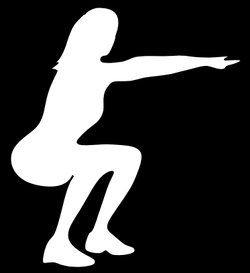
Build a Stable Foundation

It is extremely important to remember that is best to ease into training - we never add strength to dysfunction! Chances are, you have some kinetic chain imbalances causes by overactive, underactive, tight, and weak muscles. Stability Routines are all about correcting these imbalances by improving Neuromuscular Efficiency - our brains ability to fire specific muscles in specific patterns. These exercises should be performed at a very slow tempo, focusing on really using your mind to contract the targeted muscles.



Core Engagement

Core Training should become a fundamental component of your training program and something that is practiced multiple time each week. Your core is more than just your abs! The Core musculature also includes your glutes, hip flexors, and lower back. A well-rounded core training program should include static holds such as planks and v-sits, as well as dynamic stabilizing movements in various planes of motion such as Supermans, Bird-Dogs, and Dead Bugs (pictured above).



Posterior Chain Activation

The Posterior Chain is arguably the most important group of muscles for any athlete. The hamstrings, glutes, and back musculature work together to stabilize the knees, hips, and spine and facilitate a vast majority of our overall muscular endurance and force production capacity. Exercises to improve Posterior Chain Activation include squats, lunges, step ups, glute bridges, rows, and deadlifts performed in a slow, controlled, and conscious manor.





STRENGTH ENDURANCE

Muscular Endurance

This phase of training is all about improving your muscle fibers' ability to sustain contractions for longer durations and at higher intensities. Workouts in this phase should be more demanding and require you to dig a little deeper! Results can certainly be achieved with the use of gravity of alone, but access to any form external resistance will maximize results, and prepare your body for the demands of carrying additional weight on your climb!

Manipulate Variables

Going back to the Principle of Specificity, our body adapts to the demands it is faced. This is why it is critical to manipulate variables such as exercise selection, the amount of resistance uses, and number of repetitions and sets used. Like with anything in life, it is easy to fall into familiar patterns. While it is important to repeat staple exercises to continue developing major muscle groups, do you due diligence to discover new variations of exercises that will challenge smaller stabilizing muscles.

Recovery

When one begins to incorporate structured strength training, there is very high likelihood of experiencing DOMS - Delayed Onset Muscle Soreness. This is perfectly normal and a sign that your body is responding to the stimulus of training! Nonetheless, it is very important that you give your body ample time to recover before engaging in workouts that utilize the same muscle groups. Proper nutrition and mobility techniques will help expedite recovery and are explained in greater detail later in this document.







MOBILITY TRAINING



Joint Range of Motion

Improved biomechanics, muscle activation, and injury prevention are just some of the key highlights one will experience from optimizing their joint range of motion. Through repetitive motion of daily activities and recreating outside, our body can become limited to moving in various ways. This limited range of motion can decrease the neural activation of muscles and increase our chances of injury. Improving joint range of motion can be achieved by practicing various stretching and myofascial release techniques daily. Grab your foam roller!

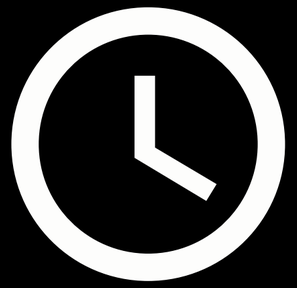




Results & Recovery

Daily mobility training is an excellent tool to expedite the recovery process and get your body ready to train again. In addition to recovery, proper joint range of motion and supple muscles allows the nervous system to fire a larger percentage of available muscle fibers, increasing the overall contractile strength of the muscle. Foam rolling is an amazing tool to break up knots and adhesions within the muscle, releasing scar tissue and allowing the entire muscle and fascia to move with greater freedom.



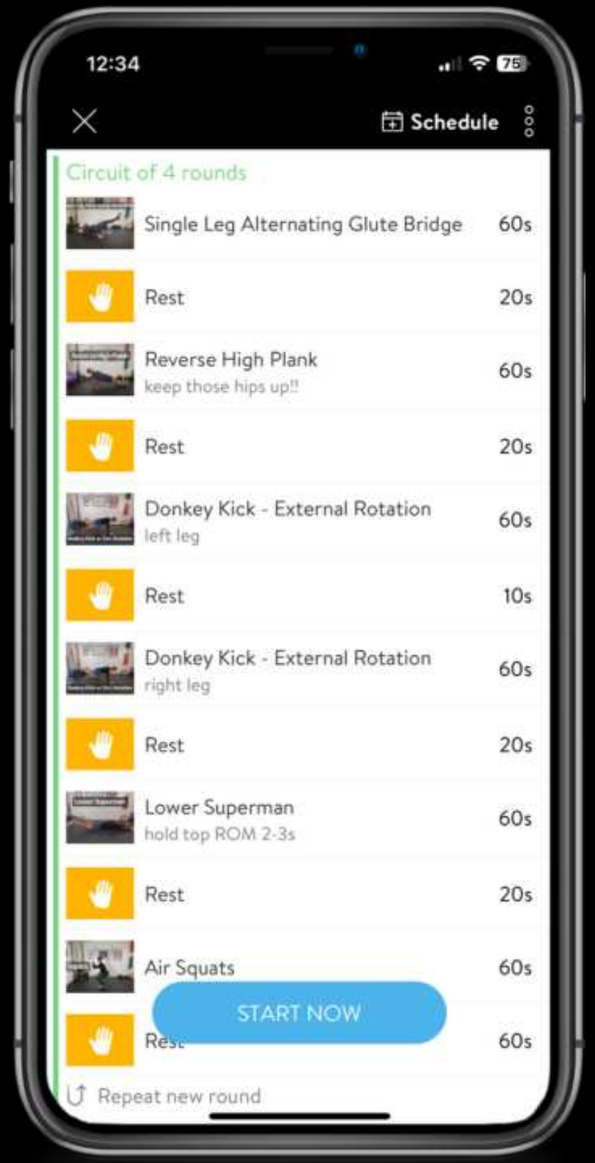


Longevity!

Truly the highlight of Mobility Training, enhancing muscle flexibility and joint range of motion is a sure-fire way to keep you moving in the outdoors for many decades to come. While this one climb you are training for is very important, be careful not to lose sight of the bigger picture of life. As we expose ourselves to the motions of training and daily life; routine stretching, foam rolling, and other modalities will help you to stay limber and mobile as you age. The older we get, the more critical this becomes!

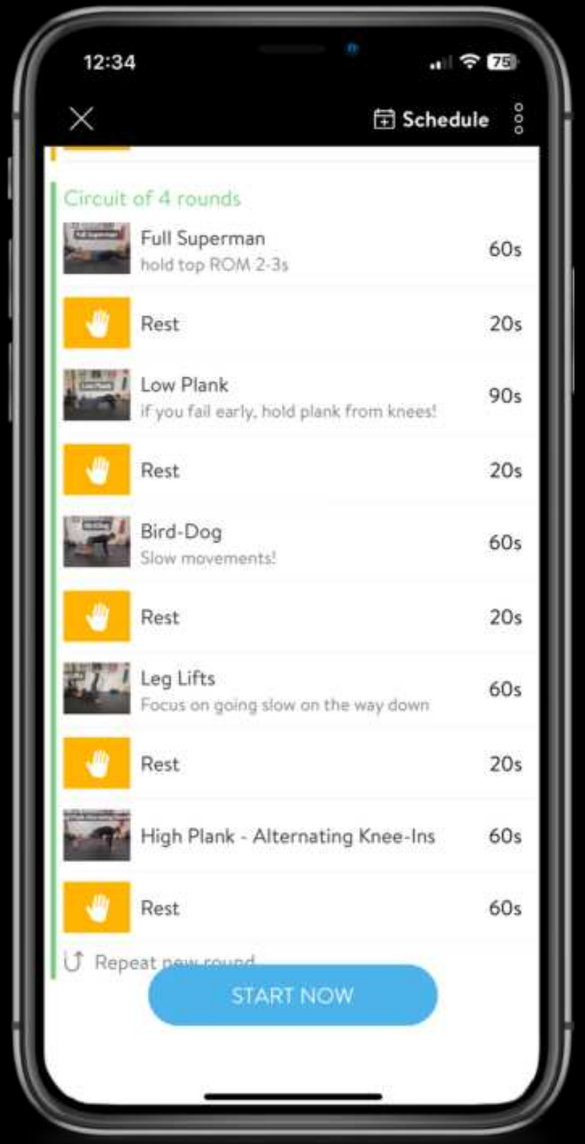
STABILITY ROUTINES

Stability Routines employ simple exercises to refine the brain-muscle connection and movement patterns!



Glute Focused Stability Routine

This workout focuses on improving posterior chain engagement by utilizing a variety of isometric and dynamic movements to activate the glutes and hamstrings. Some of these movements may feel "easy" but again, the goal is to use your BRAIN to activate the proper muscles. Translate this to your next sport-specific training objective by thinking about engaging your glutes and hamstrings during uphill travel. These exercises should be performed at a very slow and controlled tempo.

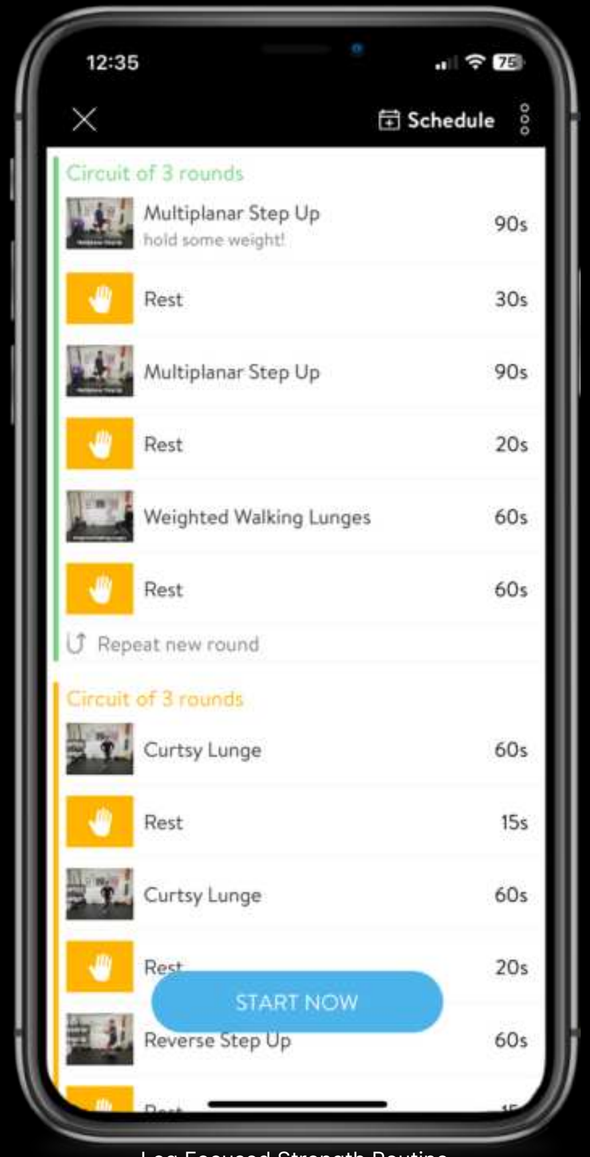


Core Focused Stability Routine

This core-focused routine uses both static and dynamic exercises to stabilize the entire core musculature. Notice how exercises generally alternate from prone to supine positions, to facilitate various sub-muscle groups within the core. Again, these exercises should be performed at an almost painstakingly slow and controlled tempo. Think quality over quantity! For example, performing only 5 Leg Lifts with excellent form is far better than 10 leg lifts performed using momentum and a disengaged core.

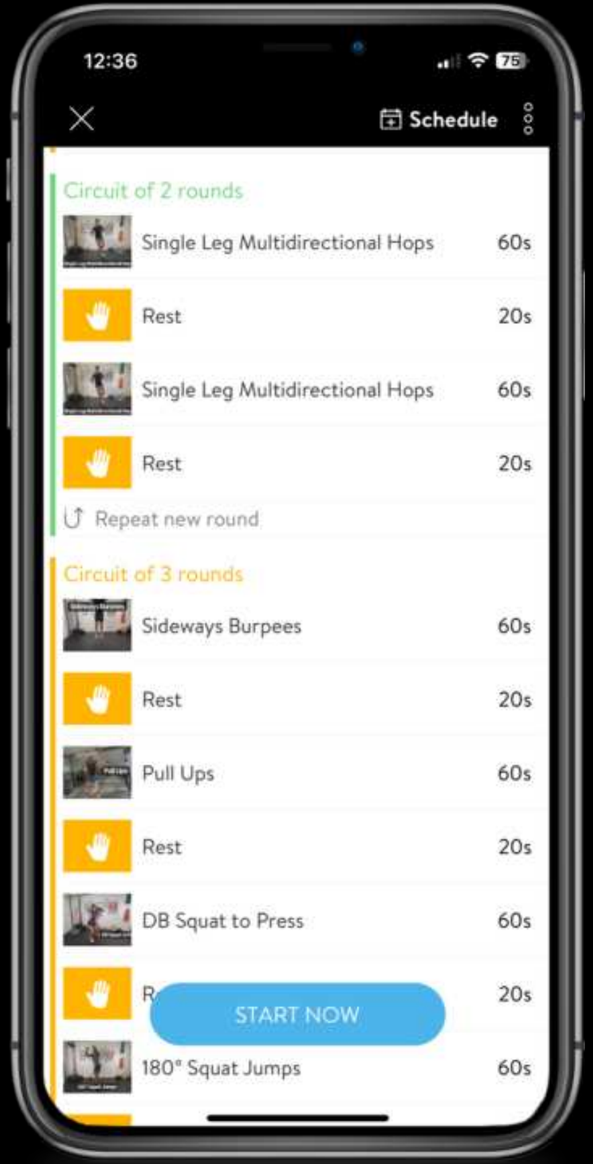
STRENGTH ROUTINES

Strength Routines can vary greatly with length of sets, rests, and duration of workout. Mix it up!



Leg Focused Strength Routine

This workout focuses on improving posterior chain strength by utilizing the lunge and step up motion. These exercises directly translate to uphill and downhill performance as they closely mimic the movement of mountain travel. Notice that the Multiplanar Step up and Curtsy Lunge require one to move in various planes of motion, requiring recruitment of smaller stabilizing muscles that are often neglected. As a general rule of thumb for step ups, start with a 10-12" box or step, and slowly progress up to 18". Don't have weights? Find a bag of rice or load up a backpack with water bottles!

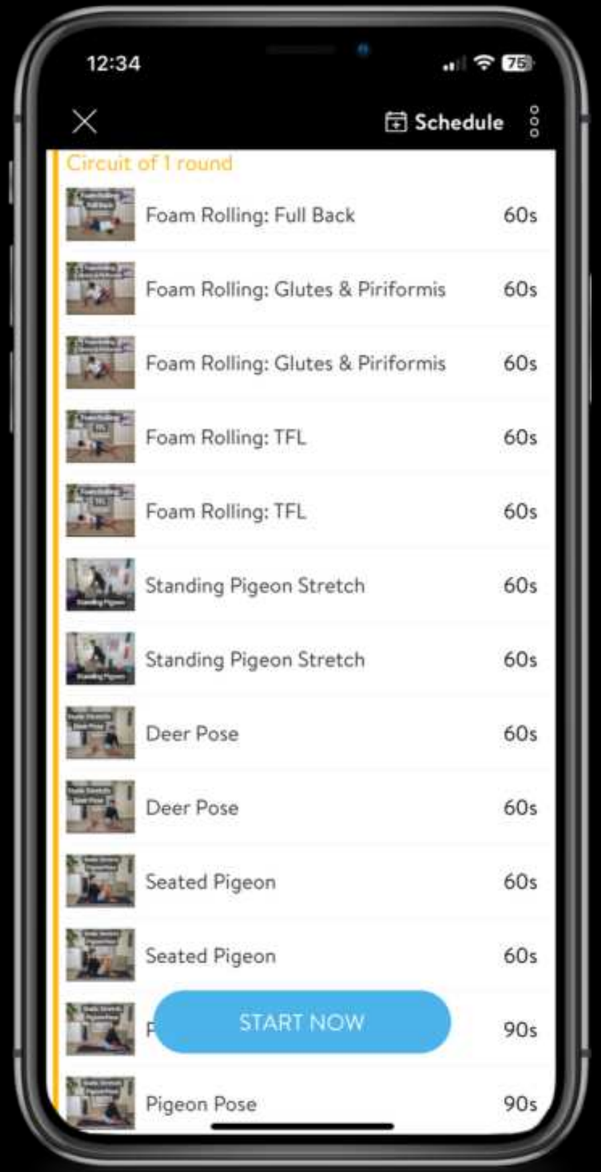


Full Body Strength Routine

This full body strength routine utilizes a wide range of exercises and training methodologies including plyometrics and power movements. Alternating between upper body and lower body focused exercises stimulates greater caloric expenditure to maximize your training hours - called Peripheral Heart Action System. Due to the higher intensity nature of exercises like Burpees and 180 Squat Jumps, the overall Rate of Perceived Exertion (RPE) should be higher and place greater demand on the involved muscular systems.

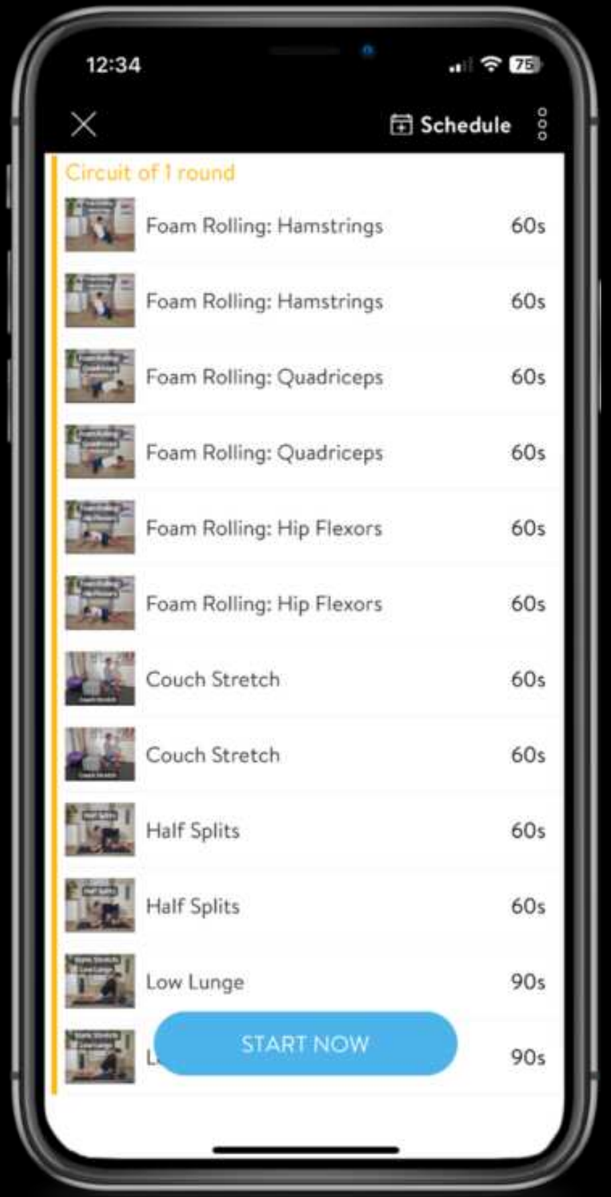
MOBILITY TRAINING

Stretching and Self-Myofascial Release should be performed DAILY to improve mobility & flexibility!



Hip Focused Mobility Routine

This mobility routine utilizes foam rolling and static stretching to improve flexibility and joint range of motion in the external rotators of the hips. Walking uphill and carrying a backpack, can put tremendous stress on the external rotators and these structures can get TIGHT! Because of this, getting deep into the hips can be difficult, physically and emotionally. Ease into these poses slowly and do your best to stay relaxed and breathe to relax your nervous systems and open up those hips!



Leg Focused Mobility Routine

Focusing on the quadriceps and hamstrings, this well-rounded mobility routine opens up the major muscles of the thighs. Repetitive use from walking up and downhill, not to mention the chronic issues created from sitting, the quads and hamstrings often become overactive (tight) in most people. This short 14-minute routine performed 3-4 times per week will improve recovery times, enhance mobility, and optimize uphill power!

6-WEEK SAMPLE PROGRAM

M	T	W	T	F	S	S
30 Lunges & Core Hip Mobility	31 Run/Hike Cardio Hamstring Mobility	1 Upper Body & Core Spine Mobility	2 Step Ups & Core Quad Mobility	3 Active Recovery	4 Long Hike Lower Leg Mobility	5 Recovery Hike Full Body Stretch
6 Squats & Core Hip Mobility	7 Run/Hike Cardio Hamstring Mobility	8 Full Body Strength Spine Mobility	9 Step Ups & Core Quad Mobility	10 Active Recovery	11 Long Hike Lower Leg Mobility	12 Recovery Hike Full Body Stretch
13 Lunges & Core Hip Mobility	14 Run/Hike Cardio Hamstring Mobility	15 Upper Body & Core Spine Mobility	16 Step Ups & Core Quad Mobility	17 Active Recovery	18 Long Hike Lower Leg Mobility	19 Recovery Hike Full Body Stretch
20 Squats & Core Hip Mobility	21 Run/Hike Cardio Hamstring Mobility	22 Full Body Strength Spine Mobility	23 Step Ups & Core Quad Mobility	24 Active Recovery	25 Long Hike Lower Leg Mobility	26 Recovery Hike Full Body Stretch
27 Lunges & Core Hip Mobility	28 Run/Hike Cardio Hamstring Mobility	1 Upper Body & Core Spine Mobility	2 Step Ups & Core Quad Mobility	3 Active Recovery	4 Long Hike Lower Leg Mobility	5 Recovery Hike Full Body Stretch
6 Squats & Core Hip Mobility	7 Run/Hike Cardio Hamstring Mobility	8 Upper Body & Core Spine Mobility	9 Step Ups & Core Quad Mobility	10 Active Recovery	11 Long Hike Lower Leg Mobility	12 Recovery Hike Full Body Stretch

Sample of Custom Summit Program from OAT

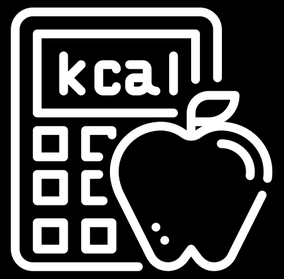
Key Points

- **At least one long hike each week is the only way to improve aerobic fitness. A "long" hike is relative to the athlete and their level training, and could range for a 3-12+ miles. Each week, do your best to slowly increase mileage, elevation gain, and pack weight!**
- **One moderate to high intensity effort each week is effective to improve your anaerobic threshold and VO2Max. Understand that these shorter efforts are helpful, but cannot replace long, slow days.**
- **Logging hikes on back-to-back days is an excellent way to prepare your body for big mountaineering efforts.**
- **Daily mobility practice is essential! This can happen in the form of foam rolling, static stretching, dynamic stretching, yoga, and massage!**
- **Switch up your strength routines! Notice the weekly variety of squats, lunges, step ups, core, upper body, and full body strength-focused workouts. A well-rounded strength program will make you a well-rounded athlete!**
- **Active Recovery is important! Take at least one or two days each week to focus on mobility, nutrition, hydrating, mental stress relief and preparing for your next day of training!**

NUTRITION

True Results Happen in the Kitchen!





Calculate Your Needs

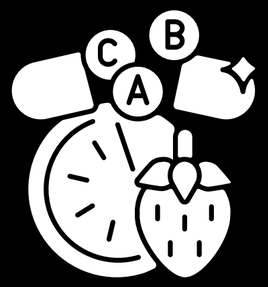
One of the most common mistakes made by many outdoor athlete is misinterpreting the correct quantity of food they need to consuming in order to perform optimally, resulting in fatigue and decreased performance. This can be mitigated by calculating your Basal Metabolic Rate and Total Caloric Needs using various online resources, such asl, Active.com. After calculating approximately how many calories you require each day, you can track your caloric intake on an app like My Fitness Pal to ensure you're getting enough energy!



Nutrient Timing

During periods of high training volumes, WHEN we eat becomes just as important as WHAT we eat! 1-2 hours before training bouts, one should consume a well-balanced meal of carbohydrates, healthy fats, and protein. During workouts lasting more than 60 minutes, be sure to consume simple, easily digestible carbs every hour or so, this is similar to how you will eat during your climb. Be sure to replenish the tank with healthy, protein-rich and carbohydrate loaded foods to expedite recovery!

Want more on Nutrition? Scan the QR Code and Download our Nutrient Timing PDF!



Micronutrients

Vitamins and Minerals deliver critical micronutrients that aid in cellular repair, immune system function, energy production, and much more! These nutrients are most often found in whole food, plant-based sources such as fruits, vegetables, whole grains, nuts, seeds, and legumes. Micronutrients are not only beneficial for training, but contribute to maintaining optimal health and disease prevention.



TRAINING AT YOUR FINGERTIPS!

Training for mountaineering can be an intimidating undertaking. We have seen so many people fall short of the summit because they simple were not physically prepare. Working with a fitness coach significant increases your chances of success!

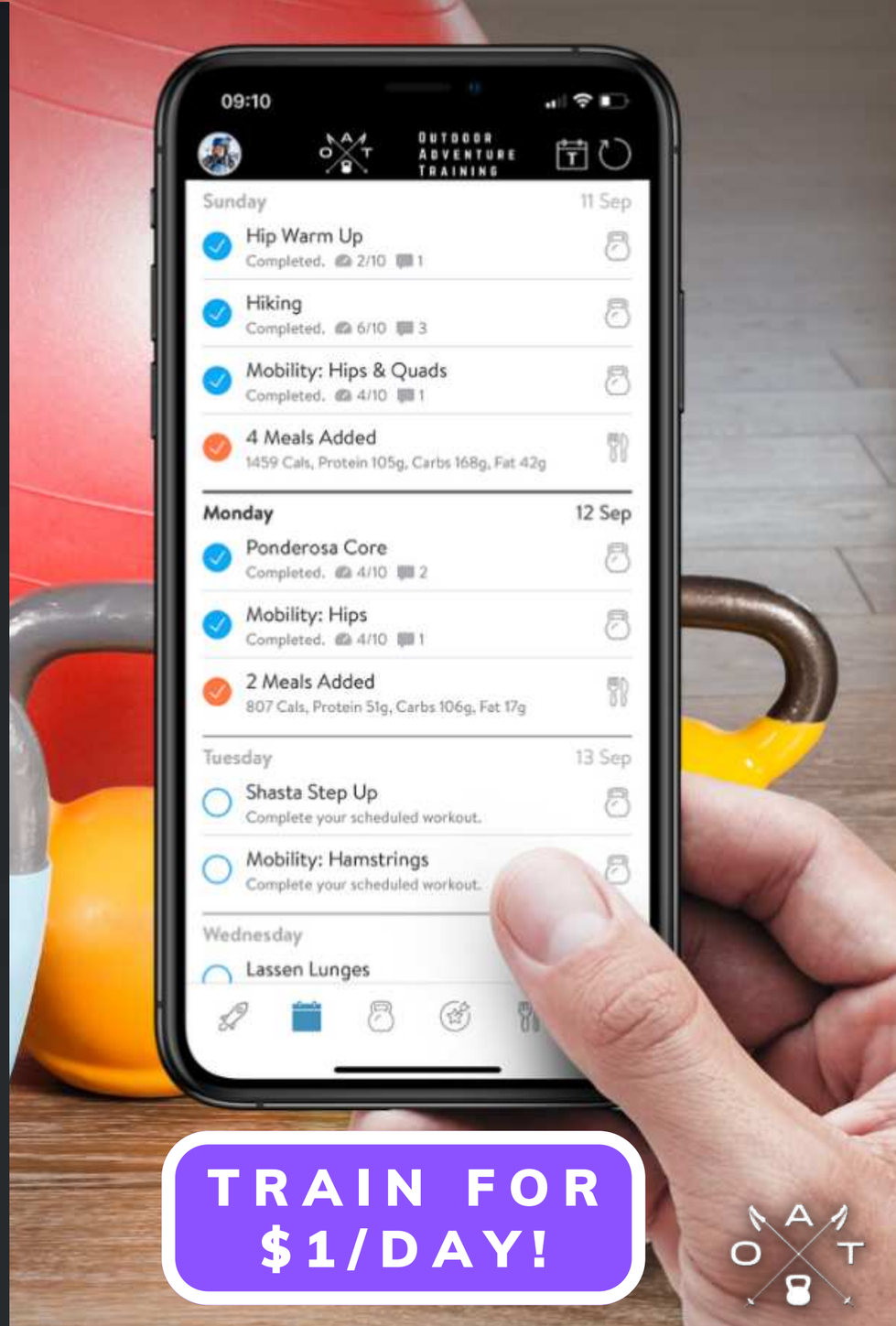
Outdoor Adventure Training offers fully remote training programs built to support outdoor athletes in reaching the summit of their dreams. Our unique mobile app loaded with tutorial videos allows for high-level remote learning and accountability to keep you on track to meet you goals.

OAT offers a variety of programs all with various levels of support. From Basecamp to Summit, we have you covered with custom training, daily workouts, mobility routines, training rationale videos, and so much more!

Scan the QR Code to Book a Consultation!

YouTube

Check out OAT's YouTube Channel for follow-along workout videos!



TRAIN FOR $1/DAY!



MIKEY BELL

TRAINER & GUIDE
OWNER OF O.A.T.

Mikey has been a Certified Personal Trainer for over 10 years, working with hundreds of clients all with different backgrounds, limitations, and goals. Complimented by 6 years of mountain guiding experience, Mikey knows what it takes to be physically and mentally prepared for big outdoor pursuits. Mikey's deepest passion in life is empowering others to reach their highest potential and accomplish their biggest dreams!

- NASM CERTIFIED PERSONAL TRAINER
- PROFESSIONAL MOUNTAIN GUIDE
- MASTERS OF OUTDOOR ADVENTURE & EXPEDITION LEADERSHIP
- CERTIFIED CORRECTIVE EXERCISE SPECIALIST
- CERTIFIED PLANT-BASED SPORTS NUTRITION COACH


SHASTA
MOUNTAIN GUIDES
SINCE 1982
QUESTIONS?
We are here to support YOU!
If you have additional questions about preparing for your climb, please reach out to Shasta Mountain Guides and Outdoor Adventure Training.
O A T

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